

Topological Set Theoretic Estimation: Follow-on Work of the Past Decade (2015–2026)

Project `thejeb`

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Abstract

A targeted survey of work in the decade since Carlson’s 2012 topological-space recast of set theoretic estimation (STE) [3]. Two findings dominate. First, the *topological*, non-convex branch of STE is, as far as we can determine, an almost single-author thread: Carlson and his direct collaborators, with little independent uptake. Second, no proof assistant has mechanized the multi-set feasibility machinery that STE sits on, which makes the present project’s Lean development first-of-kind. Claims below are attributed to primary sources; nothing here is asserted proven in our library unless it is machine-checked in `Ste`.

1 Carlson’s own program (2014–2026)

The dissertation grew into a sustained line of IEEE conference papers with a rotating co-author cluster (early on R.E. Hiromoto and R.B. Wells; later B. Ghosh, I.K. Dutta, M. Totaro, S.R. Mikkilineni, B. Williams, M. Singh, and others). The consistent thesis: reduce every cipher to an equivalent *substitution* cipher, then attack it with STE, the AEP, and unicity distance.

- Carlson et al. [4] — journal version of the dissertation’s known-ciphertext attack, with the original advisors.
- Ghosh et al. [7] — product ciphers reduce to a single substitution cipher via a “meta-alphabet” construction. This is the published counterpart of what we mechanize in `Ste.Carlson.Reduction` (Theorem 5.2, Corollary 5.3).
- Carlson et al. [1] — uses the isomorph count (the content of Carlson’s Lemma 4.1, our `card_consistent_keys`) to shrink the effective brute-force key space.
- Carlson et al. [5] — the most substantive information-theoretic follow-on: Shannon’s global unicity distance is an *average*, not a lower bound, and a per-string *local* unicity is defined. A natural (deep) future mechanization target.
- Carlson et al. [6] — security implications of the product-to-substitution equivalence.
- The mode-breaking series, culminating in Hiromoto et al. [8], “Breaking the Counter (CTR) Mode.”
- Carlson et al. [2] — a GoTech/University of Maryland technical report, the closest thing to a Carlson self-survey. It restates STE (citing Combettes 1993 and Witsenhausen–Schweppe 1968), ties STE explicitly to the AEP and unicity, and frames STE as “a precursor to the neural network.” Notably it dates Carlson’s *topological*-space STE idea to **2006**, predating the dissertation.

We were unable to verify the patent portfolio attributed to Carlson in secondary sources, and therefore cite none; this is an explicit gap.

2 Independent uptake of topological STE

We found **no evidence** of anyone outside Carlson’s immediate collaboration developing STE over topological (non-metric, non-convex) spaces in this decade. The classical Hilbert-space, convex-feasibility STE lineage (Combettes 1993; see our companion survey) continues to be cited widely, but those citations remain within the convex-optimization tradition; none pivot to the topological generalization. We record this as a genuine gap and opportunity rather than padding it with tangential citations. (Absence of evidence from targeted search is not proof of absence, but multiple queries returned nothing.)

3 STE, the AEP, and unicity after 2012

Post-2012 work tying STE to information theory — property sets as AEP typical sets, and unicity as feasibility collapse — is substantial but lives entirely inside the Carlson cluster, most explicitly in Carlson et al. [5] and Carlson et al. [2]. We found no independent work connecting STE specifically to the AEP, unicity, or decryption in this window.

4 Adjacent topological / feasibility work

Being deliberately conservative: there is essentially no adjacent literature that is genuinely “STE in spirit, topological in substance” outside Carlson’s output. Keyword-adjacent hits (e.g. constructions using connected components of a topological space for encryption) are cipher *designs*, not estimation or feasibility methods, and we do not claim an STE connection for them. We deliberately exclude persistent-homology / topological data analysis, a distinct mathematical tradition with no feasibility link to STE.

5 State of mechanization — the opening

This is the decisive input for the project’s direction.

- **Lean / Mathlib.** The single-set *Hilbert projection theorem* is formalized (`Mathlib.Analysis.InnerProduct.Projection`): the existence and uniqueness of the nearest point in a nonempty complete convex set. This is the base case of every projection method — but *no* iterative multi-set feasibility algorithm (alternating projections / POCS, Bregman cyclic projections, Douglas–Rachford) is in Mathlib.
- Li et al. [9] (Optlib) formalizes convex analysis, subgradients, proximal operators, and first-order convergence *rates* on Hilbert spaces; Verified Optimization contributors [10] (CvxLean) does convex problem *modeling*. Neither is feasibility- or projection-framed.
- **Isabelle/HOL** (AFP) has strong Hilbert-space operator theory built for quantum verification, but nothing on projection-method convergence. **Coq/Rocq** has convex polyhedra via the simplex method (finite-dimensional LP only).

Net: no proof assistant has POCS, Douglas–Rachford, Bauschke–Combettes monotone-operator feasibility, or — a fortiori — topological STE. The single-set projection theorem is exactly the base case we would build on. Our roadmap (POCS convergence over `Ste.Convex`, then the topological generalization) is therefore unclaimed ground.

Method note

This survey is exploratory and, where negative, honest about it: several of the conclusions above are *absence-of-evidence* findings from targeted search, flagged as such. As always, no mathematical claim is treated as established for this project until it is machine-checked in the `Ste` library.

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